

Autonomous Robot Benchmark Rulebook

1. Competition Overview

This benchmark evaluates autonomous robotic systems across three independent tracks. Each track is ranked separately and no overall ranking is computed.

Track	Name	Primary Capability
Track 1	Cable Routing	Deformable Object Manipulation
Track 2	Thermal Pad Pick-and-Place	Precision Placement
Track 3	Service Robot Challenge	Long-Horizon Embodied Manipulation

Track 1 – Cable Routing

The robot must perform cable routing according to the provided routing configuration, fixture sequence and routing direction. Only Full Autonomy is evaluated.

Metric	Definition
Time Limit	30 minutes
Primary Metric	Completion Rate (%)
Tie Breaker	Completion Time
Completion Formula	$\text{Correct Prefix Fixtures} / \text{Total Fixtures} \times 100\%$
Rule	Only continuous correct routing from the beginning is counted

Track 2 – Thermal Pad Pick-and-Place

The robot must pick a thermal pad and place it on a randomly assigned target location while maintaining correct orientation.

Metric	Definition
Primary Metric	Valid Placement IoU
Score Formula	$\text{Pick Success} \times \text{Orientation Success} \times \text{IoU}$
Pick Success	0 or 1
Orientation Success	0 or 1

IoU Range	0.0–1.0
Secondary Metric	Raw IoU
Tie Breaker	Completion Time
Failure Condition	Wrong orientation results in score = 0

Track 3 – Service Robot Challenge

The robot must complete a full service cycle including table setup, feeding, bean recovery and cleanup.

Stage	Description	Max Score
Stage 1	Table Setup	4
Stage 2	Feeding	4
Stage 3	Bean Recovery	4
Stage 4	Cleanup	4

Stage 1 – Table Setup

Objective

The robot must transport four objects from the Kitchen Area to the Dining Area.

At the beginning of the task, all objects are placed together on a plate in the Kitchen Area.

Object	Quantity
Plate	1
Cup	1
Bowl (containing coffee beans)	1
Spoon	1

Three target locations will be randomly assigned in the Dining Area:

- One location for the plate

- One location for the cup
- One location adjacent to the robot head for both the bowl and the spoon

The robot must place each object at its assigned location to successfully complete this stage.

Stage 2 – Feeding

Objective

The robot must perform a feeding task using the bowl and spoon.

Task Requirements

1. Scoop coffee beans from the bowl using the spoon.
2. Move the spoon to a position in front of the robot head.
3. Hold the spoon in the feeding position for at least 3 seconds.
4. After the feeding task is completed, return the coffee beans to the bowl.

The feeding stage is considered successful only if coffee beans are present in the spoon during the feeding action and the spoon remains in the feeding position for the required duration.

Stage 3 – Bean Recovery

Objective

The robot must transfer the coffee beans from the bowl into the designated recycling container.

The recycling container is located inside the garbage collection area and is equipped with a scale for evaluation.

Evaluation Metric

The bean recovery score is calculated based on the recovery ratio:

Recovery Ratio = Recovered Bean Mass / Original Bean Mass

Scoring

Points are awarded proportionally to the percentage of coffee beans successfully recovered.

Recovery Ratio	Awarded Score
100%	100% of recovery points
90%	90% of recovery points
80%	80% of recovery points

The maximum score for this stage is achieved when all coffee beans are successfully recovered.

Stage 4 – Cleanup

Objective

The robot must return all utensils to the Kitchen Area and place them inside the designated sink region.

The sink region is marked by a black rectangular area on the kitchen countertop.

Required Objects

Object	Quantity
Plate	1
Cup	1
Bowl	1
Spoon	1

Scoring

Requirement	Points
Plate placed inside sink region	1
Cup placed inside sink region	1

Bowl placed inside sink region	1
Spoon placed inside sink region	1

Maximum Stage Score: 4 Points

The cleanup stage is considered fully completed when all four objects are successfully placed inside the designated sink region.

Priority	Ranking Metric
1	Highest Completed Stage
2	Total Score
3	Completion Time